

2.

Accordingly, the answers of selecting item i may result in different values, yes or no?

Ans: Yes, Different selections of previous items can lead to different results.

Consequently, can we memoize this brute-force code for speed-up?

Ans: No. The function only tracks the item index and ignores the previous item selections, so it cannot be memoized correctly.

5. Observe. Is each state generated by the brute-force algorithm unique? In other words, are there repeated occurrences of the same state ?

Ans: No. They are not unique, and there are repeated occurrences of the same state because the same item index and remaining capacity can be reached through different recursive paths.

6. Given a problem state, does any recursive call on this state return the same value? Why?

Ans: Yes. Every recursive call on the same state returns the same value because the remaining items and remaining capacity are identical, regardless of the recursive path taken to reach that state.

8. Compare the total number of recursive calls against the brute-force version.

Ans: The memoized version makes significantly fewer recursive calls than the brute-force version because each state is computed only once and then reused.